THAMISH WANDURAGALA

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ABOUT ME

I am a first-year ECE PhD student interested in cyber-physical systems and hardware–software co-design, with a background in computer architecture, hardware design, and low-level programming. My goal is to develop efficient, system-level solutions that integrate physical processes with computing platforms.

EDUCATION

University of Maryland, College Park, USA **2025 August - Present**

- PhD student in Electrical and Computer Engineering — Current GPA: 4.00/4.00

University of Peradeniya, Sri Lanka **2019 November - 2024 August**

- B.Sc. (Eng.) First Class Honors in Computer Engineering — GPA: 3.95/4.00

Trinity College, Kandy, Sri Lanka **2004 - 2018**

- G.C.E. Advanced Level (2018) - Physical Science Stream.
- G.C.E. Ordinary Level (2015)

EXPERIENCE

Visiting Research Assistant **2023 August - 2024 January**

- Advised by Prof. Yuen Chau at Engineering Product Development Pillar, Singapore University of Technology and Design (SUTD), Singapore in collaboration with COMSO Lab, Nanyang Technological University (NTU).

Teaching Assistant **2025 August - Present**

- Department of Electrical and Computer Engineering, University of Maryland, USA
  - ENEE446: Digital Computer Design

Teaching Assistant **2024 October - 2025 August**

- Department of Computer Engineering, Faculty of Engineering, University of Peradeniya, Sri Lanka
  - CO502 Advanced Computer Architecture
  - CO3060 Computer Systems Design Project

Casual Instructor **2024 January - August**

- During my senior year, supervised weekly sessions for freshmen, sophomores for course modules, Operating Systems, Signal Processing, Programming Methodology (Based on C Programming Language), Digital Design

RESEARCH

Neuromorphic Architecture for Spiking Neural Networks *(Group)*| *Verilog* **2023 Dec - 2025 Jan**

- A neuromorphic accelerator architecture that is designed to run small scale Spiking Neural Networks (SNNs), using a network of nodes optimised for high speeds and low power consumption.
- The synchronous accelerator supported 32-bit arithmetic operations, enhancing the inferencing capabilities of the SNN. Testing was conducted on an Altera FPGA, and performance was evaluated using Synopsys tools, including PrimePower and PrimeTime.
- Contributed to designing the architecture and testing its performance.
- Manuscript 1, Manuscript 2.

2-bit Intelligent Reflective Surface Prototype *(Group)* | *Python, Matlab* **2023 August - 2024 January**

- A 6G Wireless Communication research project focused on Reflective Intelligent Surfaces (RIS).
- Developed a hardware controller for the wireless device, built the testing platform, tested the device for various applications including beam tracking.

PROJECTS

Automated Road-Rule Detector *(Group)* | *Python, Flutter, Firebase, Nodejs* **2022 October - 2023 January**

* A hardware device based on Raspberry Pi 3 as the microprocessor which is able to detect road rule breaking severity with an app to visualize collected data.
* Contributed to the hardware development and cloud deployment.

8-bit Single Cycle Processor *(Group)* | *Verilog* **2021 November - 2022 March**

* An 8- bit single cycle processor with an ALU, register file, control logic, data memory, data cache, instruction memory and instruction cache. A memory hierarchy too was built.

ML-based Patient Diagnosis/Care *(Group)* | *Python, Nodejs, Flutter* **2023 May - 2023 July**

* An app designed to get symptoms from a user and provide insights and suggestions on potential risks, recommended eating habits and recommendations on type of medical doctor to visit.
* Contributed to the Machine Learning model and the Back-End.

Water Quality Monitoring System *(Group)* | *ESP8266, SCADA* **2023 May - 2023 July**

* A water quality monitoring system using ESP8266 microcontroller, which continuously monitors the temperature and turbidity of water in real-time.
* Contributed to the development of hardware and the deployment of the SCADA.

Smart Inventory Management System *(Group)* | *Laravel with php, blade, MYSQL* **2022 May - August**

* Introduced a resource scheduling system to manage resources in the MakerSpace Lab (Department of Computer Engineering, University of Peradeniya).

Database System at University Gymnasium *(Group)* | *MySQL, Php* **2021 November - 2022 March**

* A fully functional database to organize reservations, cancellations, borrowings and misplacements of equipment at the university gym.

ACHIEVEMENTS / EXTRACURRICULAR

Global Robotics Games *(Group)* **November 2023**

- 1st place in University Category - Three day competition based on developing robotic controllers for Transformer Robot Soccer League.

Aces Hackathon 2023 *(Group)* **June 2023**

- 2nd place in "Other" category - Two day competition with 40+ participating teams.

Aces Pre-Coders 2022 *(Group)* **August 2021**

- Finalists - A 6 hour coding competition with 70+ participating teams.

Offices Held

- President, Ceylon University Dramatic Society, University of Peradeniya. (2022-2023)
- Secretary, Aviation Society, Trinity College, Kandy. (2017-2018)

Sports

- Baseball Age categories 19, 17, 15, 13 - Trinity College, Kandy. (2011 - 2017)

SKILLS

Hardware Prototyping: Verilog, Synopsys Primepower, DC-Compiler, ARM Assembly, Quartus
Programming Languages: Python, Java, C, C#

Mobile Development: Flutter, Native Android

Front End: HTML/CSS, Bootstrap

Back End: Spring Boot, Nodejs

Data Modeling:Numpy, Pandas, scipy, sklearn

Databases: MySQL, FireBase

Version Controlling: Git

REFEREES

Prof. Chau Yuen |**chau.yuen@ntu.edu.sg**
Associate Professor,
School of Electrical and Electronic Engineering,
Nanyang Technological University (NTU),
Singapore

Prof. Roshan G. Ragel |**roshanr@eng.pdn.ac.lk**
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